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*                                                *
*           STAAD.Pro V8i SELECTseries3          *
*           Version  20.07.08.20                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUN 18, 2012                *
*           Time=    11: 2:38                    *
*                                                *
*      USER ID: RNS Design _Engineering         *
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     1. STAAD SPACE
INPUT FILE: 20120618_ Air Compr Shed_r1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 29-SEP-11
     4. JOB NAME JRC-AREA 198 FOAM CONDENSATE SHED
     5. JOB PART AIR COMPRESSOR SHED
     6. END JOB INFORMATION
     7. INPUT WIDTH 79
     8. UNIT METER KN
     9. JOINT COORDINATES
    10. 1 0 0 0; 2 6 0 0; 3 0 0 7; 4 6 0 7; 5 0 4.5 0; 6 6 4.5 0; 7 0 5.1 7; 8 6 5.1 7
    11. 11 0 4.4 -1; 13 6 4.4 -1; 14 0 5.2 8; 16 6 5.2 8; 19 -1 4.4 -1; 20 -1 5.2 8
    12. 23 7 4.4 -1; 24 7 5.2 8; 25 0 4.58571 1; 26 0 4.67143 2; 27 0 4.75714 3
    13. 28 0 4.84286 4; 29 0 4.92857 5; 30 0 5.01429 6; 37 6 4.58571 1; 38 6 4.67143 2
    14. 39 6 4.75714 3; 40 6 4.84286 4; 41 6 4.92857 5; 42 6 5.01429 6
    15. 43 -1 4.58571 1; 44 -1 4.67143 2; 45 -1 4.75714 3; 46 -1 4.84286 4
    16. 47 -1 4.92857 5; 48 -1 5.01429 6; 49 7 4.58571 1; 50 7 4.67143 2
    17. 51 7 4.75714 3; 52 7 4.84286 4; 53 7 4.92857 5; 54 7 5.01429 6
    18. 55 -1 4.48955 -0.104466; 56 7 4.48955 -0.104466; 57 -1 5.10995 7.0995
    19. 58 7 5.11 7.0995; 59 0 5.10995 7.0995; 60 6 5.11 7.0995
    20. 62 0 4.48955 -0.104466; 64 6 4.48955 -0.104466
    21. MEMBER INCIDENCES
    22. 1 1 5; 2 3 7; 3 2 6; 4 4 8; 6 5 25; 9 6 37; 10 5 6; 12 7 8; 15 11 62; 16 7 59
    23. 19 13 64; 20 8 60; 31 19 11; 32 11 13; 34 13 23; 35 20 14; 36 14 16; 38 16 24
    24. 39 25 26; 40 26 27; 41 27 28; 42 28 29; 43 29 30; 44 30 7; 51 37 38; 52 38 39
    25. 53 39 40; 54 40 41; 55 41 42; 56 42 8; 69 43 25; 70 25 37; 72 37 49; 73 44 26
    26. 74 26 38; 76 38 50; 77 45 27; 78 27 39; 80 39 51; 81 46 28; 82 28 40; 84 40 52
    27. 85 47 29; 86 29 41; 88 41 53; 89 48 30; 90 30 42; 92 42 54; 93 5 8; 94 7 6
    28. 103 55 62; 106 57 59; 107 59 14; 108 60 16; 109 60 58; 110 59 60; 113 62 5
    29. 115 64 6; 116 64 56; 117 64 62
    30. DEFINE MATERIAL START
    31. ISOTROPIC STEEL
    32. E 2.05E+008
    33. POISSON 0.3
    34. DENSITY 76.8195
    35. ALPHA 1.2E-005
    36. DAMP 0.03
    37. END DEFINE MATERIAL
    38. MEMBER PROPERTY BRITISH
    39. 1 TO 4 6 9 10 12 15 16 19 20 39 TO 44 51 TO 56 107 108 113 -
    40. 115 TABLE ST UC254X254X73
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31 32 34 TO 36 38 69 70 72 TO 74 76 TO 78 80 TO 82 84 TO 86 88 TO 90 92 103 -
106 109 110 116 117 TABLE ST CH150X75X18
93 94 TABLE ST UA60X60X6
CONSTANTS
BETA 90 MEMB 1 TO 4
BETA 355 MEMB 31 34 35 38 69 72 73 76 77 80 81 84 85 88 89 92 103 106 109 116
BETA 45 MEMB 93 94
MATERIAL STEEL ALL
SUPPORTS
1 TO 4 PINNED
************************************
*DEAD LOAD
************************************
MEMBER TENSION
93 94
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
MEMBER LOAD
*(DL=0.25KN/M2 X 1 = 0.25 KN/M)
69 70 72 TO 74 76 TO 78 80 TO 82 84 TO 86 88 TO 90 92 103 106 109 110 116 -
117 UNI GY -0.25
*(DL=0.25KN/M2 X 0.5 = 0.125 KN/M)
31 32 34 TO 36 38 UNI GY -0.125
************************************
*LIVE LOAD
************************************
LOAD 2 LIVE LOAD
MEMBER LOAD
*(LL=0.75KN/M2 X 1 = 0.75 KN/M)
69 70 72 TO 74 76 TO 78 80 TO 82 84 TO 86 88 TO 90 92 103 106 109 110 116 -
117 UNI GY -0.75
31 32 34 TO 36 38 UNI GY -0.375
*(LL=0.75KN/M2 X 0.5 = 0.38 KN/M)
************************************
*WIND LOAD - X DIR
************************************
LOAD 3 WIND LOAD X-DIR
MEMBER LOAD
1 TO 4 6 15 16 39 TO 44 107 113 UNI GX 0.22
*(WIND LOAD WL=0.85KN/M2 X 0.41 = 0.35 KN/M)
************************************
*WIND LOAD - Z DIR
************************************
LOAD 4 WIND LOAD Z-DIR
MEMBER LOAD
1 TO 4 UNI GZ -0.22
35 36 38 UNI GZ -0.22
*(WIND LOAD WL=0.85KN/M2 X 0.41 = 0.35 KN/M)
************************************
*NOTIONAL LOAD - X DIR
************************************
LOAD 5 NOTIONAL LOAD X-DIR
SELFWEIGHT X 0.015
MEMBER LOAD
69 70 72 TO 74 76 TO 78 80 TO 82 84 TO 86 88 TO 90 92 103 106 109 110 116 -
117 UNI GX 0.00375
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 97. 31 32 34 TO 36 38 UNI GX 0.001875
 98. ************************************
 99. *NOTIONAL LOAD - Z DIR
100. ************************************
101. LOAD 6 NOTIONAL LOAD Z-DIR
102. SELFWEIGHT Z -0.015
103. MEMBER LOAD
104. 69 70 72 TO 74 76 TO 78 80 TO 82 84 TO 86 88 TO 90 92 103 106 109 110 116 -
105. 117 UNI GZ -0.00375
106. 31 32 34 TO 36 38 UNI GZ -0.001875
107. ************************************
108. *SERVICE LOAD
109. ************************************
110. LOAD COMB 7 (1.0DL+1.0LL)
111. 1 1.0 2 1.0
112. LOAD COMB 8 (1.0DL+1.0LL+1.0WLX)
113. 1 1.0 2 1.0 3 1.0
114. LOAD COMB 9 (1.0DL+1.0LL+1.0WLZ)
115. 1 1.0 2 1.0 4 1.0
116. LOAD COMB 10 (1.0DL+1.0LL+1.0NLX)
117. 1 1.0 2 1.0 5 1.0
118. LOAD COMB 11 (1.0DL+1.0LL+1.0NLZ)
119. 1 1.0 2 1.0 6 1.0
120. ************************************
121. *ULTIMATE LOAD
122. ************************************
123. LOAD COMB 12 (1.4DL+1.6LL)
124. 1 1.4 2 1.6
125. LOAD COMB 13 (1.2DL+1.2LL+1.2WLX)
126. 1 1.2 2 1.2 3 1.2
127. LOAD COMB 14 (1.2DL+1.2LL+1.2WLZ)
128. 1 1.2 2 1.2 4 1.2
129. LOAD COMB 15 (1.2DL+1.2LL+1.2NLX)
130. 1 1.2 2 1.2 5 1.2
131. LOAD COMB 16 (1.2DL+1.2LL+1.2NLZ)
132. 1 1.2 2 1.2 6 1.2
133. PERFORM ANALYSIS
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P R O B L E M   S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =   48/   60/   4

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

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ORIGINAL/FINAL BAND-WIDTH=    42/    11/     69 DOF
TOTAL PRIMARY LOAD CASES =    6, TOTAL DEGREES OF FREEDOM =    276
SIZE OF STIFFNESS MATRIX =       20 DOUBLE  KILO-WORDS
REQRD/AVAIL. DISK SPACE  =     12.3/   7428.3 MB
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**START ITERATION NO.   2